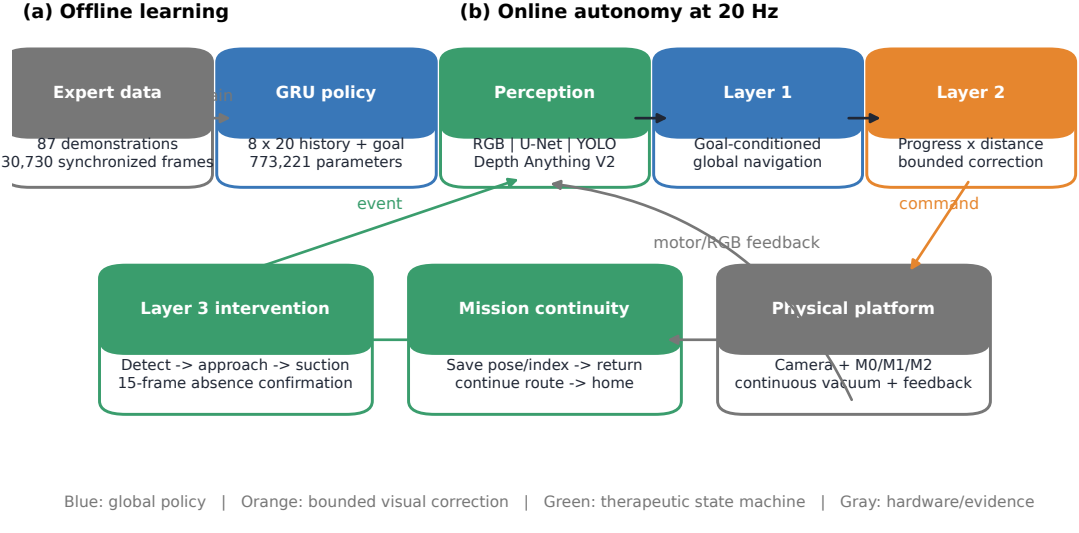
(a) Offline learning
(b) Online autonomy at 20 Hz
Expert data
87 demonstrations
30,730 synchronized frames
train
GRU policy
8 x 20 history + goal
773,221 parameters
Perception
RGB | U-Net | YOLO
Depth Anything V2
Layer 1
Goal-conditioned
global navigation
Layer 2
Progress x distance
bounded correction
event
command
motor/RGB feedback
Layer 3 intervention
Detect -> approach -> suction
15-frame absence confirmation
Mission continuity
Save pose/index -> return
continue route -> home
Physical platform
Camera + M0/M1/M2
continuous vacuum + feedback
Blue: global policy | Orange: bounded visual correction | Green: therapeutic state machine | Gray: hardware/evidence